

Architectural Evolution and Strategic Optimization for Room Controller Levels Three and Four

The transition through Room Controller Level (RCL) 3 and RCL 4 represents the most volatile and strategically significant pivot point in the lifecycle of a decentralized Screeps artificial intelligence. During this phase, the system moves from the "primitive survival" state—characterized by high-volatility energy income and simple, general-purpose creep roles—to an "industrialized" state defined by centralized logistics, algorithmic architectural planning, and sophisticated defensive layers. For an operator utilizing a kernel-based architecture with an Overlord-type hierarchical control structure, these levels necessitate a fundamental shift in process prioritization, moving away from simple headcount management toward specialized part-count balancing and high-efficiency resource pipelines.¹

The Technical Threshold: Infrastructure and Capacity Expansion

The fundamental constraints of a colony are governed by the Room Controller Level, which serves as the primary gating mechanism for economic throughput and defensive capability. At RCL 3, the system gains access to a single Tower and ten Extensions, increasing the total energy capacity for spawning from 550 (at RCL 2) to 800 energy units.³ This jump is significant not merely for the increased capacity but for the shift in "spawning velocity"—the ability of the system to generate more complex body-part ratios in a single creep, allowing for the deployment of specialized units such as static miners and high-capacity haulers.²

The progression to RCL 4 represents a tripling of the energy investment required at RCL 3, jumping from 135,000 to 405,000 units of energy.³ This stage introduces the Storage structure and doubles the extension count to twenty, bringing the room's total spawn energy capacity to 1,300. The emergence of the Storage structure is the singular most transformative event in the mid-game. It allows the system to transition from a "push-based" logistics model, where harvesters must find immediate sinks for energy to avoid decay, to a "pull-based" model, where energy is buffered in a central repository with a capacity of 1,000,000 units.⁴

Comparative Structural Capability and Energy Gating

Room Controller Level	Energy to Progress	Extension Count	Max Extension Capacity	Total Spawn Energy	Primary Strategic Unlock
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RCL 1	200	0	0	300	Initial Spawn / Roads
RCL 2	45,000	5	250	550	Walls / Ramparts
RCL 3	135,000	10	500	800	Tower / First Active Defense
RCL 4	405,000	20	1,000	1,300	Storage / Logistic Hub
RCL 5	1,215,000	30	1,500	1,800	Links / Energy Transmissio n

The architectural implications of these unlocks are profound. At RCL 3, the "Tower" provides a global reach defense, but its efficacy is highly dependent on its distance from the target. Damage, healing, and repair capabilities decrease linearly as the distance from the tower increases, reaching a minimum at twenty tiles or greater.¹ Consequently, the "Architect" process must prioritize tower placement that maximizes coverage of entry points and critical infrastructure while minimizing distance-based falloff.⁷

Kernel Architecture and Process Prioritization

A kernel-based architecture functions as a task-scheduling engine, managing the execution of various "Overlords"—high-level processes responsible for specific room objectives. In an Overlord/Kernel model, the tick lifecycle is divided into distinct phases: build, init, and run.⁹ This separation of concerns allows the AI to make high-level strategic decisions during the build phase, register resource and spawn requests during the init phase, and execute micro-management logic during the run phase.⁹

Process Hierarchy and Priority Queuing

In the mid-game, the kernel must balance competing demands for energy and CPU. Spawning is the highest priority when the population is below the survival threshold, but as the room matures, the priority shifts toward economic optimization and defensive fortification. The

"Overseer" class tracks "Directives" and "Overlords," running them by priority to ensure that the most critical needs are met first.⁹

The hierarchy of tasks in a healthy RCL 3 and 4 colony is typically structured to ensure that survival and economic inputs are never compromised. The "Emergency Miner" sits at the top of the priority list, triggered only when no active miners are present at a source, to prevent a "cold-boot" scenario where the colony has no energy and no creeps to harvest it.¹

Standardized Kernel Task Priorities

Priority Level	Overlord/Process	Trigger Condition	Operational Goal
0 (Critical)	Emergency Miner	Zero miners active	Restart energy cycle
1 (High)	Tower Refiller	Tower energy < 50%	Ensure room security
2 (High)	Extension Filler	Spawn/Extensions < 100%	Enable specialized spawning
3 (Medium)	Static Source Miner	Container at source < 20%	Maximize input (10 e/t)
4 (Medium)	Builder	Construction sites exist	Expand room infrastructure
5 (Low)	Controller Upgrader	Storage buffer > 100,000	Progress GCL and RCL
6 (Low)	Wall/Rampart Repairer	Hit points < Target threshold	Long-term passive defense

The "Logistics Manager" plays a central role in this hierarchy. It monitors energy levels across containers and structures, issuing transfer requests that the Task Supervisor then assigns to available Haulers.² At RCL 3, the Logistics Manager must prioritize the Tower, as it is the primary defense against NPC Invaders and early player harassment.¹

Economic Transformation: The Logistics of the

Storage Transition

The transition from RCL 3 to RCL 4 is fundamentally a shift in how the system handles energy flux. At RCL 3, the economy is "container-centric." Harvesters (Miners) sit on containers placed at energy sources and drop their harvested resources directly into them, preventing the

12.5% decay that occurs when energy sits on the ground.² These containers are limited to 2,000 units, which is relatively small given that a single source being harvested at

10 energy/tick will fill a container in 200 ticks if no haulers are present.¹¹

The Storage Buffer and Economic Smoothing

Upon reaching RCL 4 and constructing the Storage, the Logistics Manager reconfigures the hauling network to use the Storage as the primary "hub." This serves several critical functions within a kernel-based architecture. First, it significantly improves CPU efficiency by centralizing energy targets. Workers, such as Builders and Upgraders, no longer need to check multiple containers; they simply path to the Storage, reducing the number of find calls and pathfinding calculations.⁵

Second, the Storage enables "Spawning Velocity." Having a massive buffer allows the Spawn Supervisor to instantly refill extensions from the Storage, enabling the continuous spawning of large creeps without waiting for new energy to be harvested.⁵ Finally, the Storage acts as a "low-pass filter," absorbing spikes in income—such as those from remote mining, recycling dead creeps, or tombstone harvesting—and providing a steady supply for the Controller Upgrader Overlord.⁵

Storage vs. Container Logic and API Handling

A technical detail often overlooked by new players is the distinction between energy-specific structures and generic store structures. While Extensions and Spawns have a .energy and .energyCapacity property, the Storage and Container use the .store object.¹⁴ The Logistics Manager must be programmed to handle store to correctly calculate available space and current inventory.¹⁴

Structure	Property Access	Capacity	Resource Type
Extension	.energy / .energyCapacity	50 (RCL 2-6)	Energy Only
Container	.store	2,000	Multi-resource

Storage	.store	1,000,000	Multi-resource
Terminal	.store	300,000	Multi-resource

The "Architect" process must place the Storage strategically. Ideal placement is equidistant from the two energy sources and within range of the Spawner, often as part of a "core" or "bunker" layout to minimize the movement cost of the "Queen" or "Manager" creep that refills the spawn.⁸

Algorithmic Base Planning: Architecture as a Service

The "Architect" is a specialized process within the Overlord framework that runs infrequently to save CPU, studying the room terrain and deciding on optimal structure placement.² For automated bots, this usually involves one of three paradigms: bunkers, stamps, or dynamic pattern generation.⁷

Distance Transform and Open Space Identification

To automate base planning, the Architect utilizes the "Distance Transform" algorithm. This algorithm processes the room's terrain data to score each tile based on its distance from the nearest wall.⁸ A high Distance Transform score indicates a large, open area. Bunkers, which are single, large pre-defined objects (often 13×13 or 15×15), typically require a Distance Transform score of 7 or higher to be placed effectively.⁸

Stamps are smaller, modular components—such as a $5 \times$ "Fast Filler" for extensions or a "Controller Upgrade Depot"—that can be placed dynamically.⁷ This approach is more flexible in rooms with dense terrain or large swamp areas. The Architect identifies a starting position close to both the controller and the energy sources, then uses a "Floodfill" algorithm to categorize tiles by their accessibility from this core.⁸

Strategic Infrastructure Placement

Stamp/Module	Typical Size	Recommended Placement	Purpose
Core	$5 \times$	High DT score, central	Spawn, Storage, Terminal
Fast Filler	$5 \times$	Near Spawn	Rapid extension

			refilling
Upgrade Depot	3 ×	Within 3 tiles of Controller	Efficient controller upgrading
Lab Cluster	4 ×	Near Core	Compound production
Tower Cluster	Variable	Near Core/Entry points	Concentrated defensive fire

The Architect also lays down the road network. Roads are critical for increasing hauler throughput and reducing the time-to-live cost of creeps.² A common strategy is to generate roads from the Storage to energy sources, the controller, and room exits. Some bots use a dynamic road placement system where creeps drop road construction sites as they step off existing paths, eventually creating a bulky but effective network.¹⁸

Defensive Posture: Surviving the RCL 4 "Heavy" Threshold

The defensive logic of the Overlord system must undergo a significant upgrade at RCL 4 due to the change in NPC Invader behavior. The game engine tracks the total energy harvested in a room; approximately every 100,000 units mined, a new game-controlled creep appears with the goal of hunting the player's units.¹⁹

The Evolution of Invader Threat

At RCL 3 and below, invaders are "Light" creeps that are generally easy for a single tower to dispatch. However, at RCL 4, invaders transition to "Heavy" classes, which can include boosted components and squad logic.¹⁹

Invader Characteristic	RCL 3 (Light)	RCL 4+ (Heavy)
Typical Squad Size	Solo	2 to 5 creeps
Primary Roles	Melee / Ranged	Melee, Ranged, Healer
Max Hits	Base	Often Boosted

Healing Capacity	Minimal	Up to 600 hits/tick
Behavior	Aggressive	Kiting / Squad Cohesion

A single Tower at RCL 4 deals **600** damage within 5 tiles, but this damage falls off to **150** at range 20.¹ A boosted "Heavy Healer" can negate this damage entirely, allowing a squad to tank tower fire while dismantling the base.²⁰ This "imbalance" is intended to force the player to move beyond passive tower defense and implement active counter-squads.²⁰

Defensive Overlord Logic for RCL 4

The Defensive Overlord must implement sophisticated targeting and spawning logic to counter these threats. Priority must be given to Healers first, then Ranged Attackers, then Melee attackers.²⁰ If a tower is unable to kill a healer within a certain number of ticks, the Overseer should trigger an "Active Defense" directive, spawning a dedicated defender creep.⁹

Defender Role	Body Part Ratio	Combat Logic
Guard	ATTACK, MOVE	Melee engagement inside ramparts
Kiter	RANGED_ATTACK, MOVE	Kiting at range 3
Healer Support	HEAL, MOVE	Sustaining defenders during siege

Furthermore, the "Mincut" algorithm is often used by the Architect to determine the optimal positions for ramparts. This algorithm finds the minimum number of tiles needed to "shrink-wrap" the base, providing protection from ranged attacks (range 3) while minimizing the energy required for repair.⁸

Passive Defense: Wall and Rampart Maintenance

Ramparts and Walls are the primary layers of passive defense. While walls block all movement, ramparts allow the player's creeps to pass through and provide protection to structures on the same tile.²¹ Both are built with only 1 hit point and must be fortified by worker creeps.²¹

Fortification Targets and Decay

The maximum hit points for ramparts are gated by the RCL. At RCL 3, the limit is 1,000,000 hits;

at RCL 4, it increases to 3,000,000 hits.³ Maintaining these structures is an energy-intensive process, as ramparts lose 300 hit points every 100 ticks.²⁴

The "Builder Manager" must implement a prioritization system for repairs to prevent workers from "sitting" on a single high-HP wall while others decay. A common approach is to repair all structures to a minimum threshold (e.g., 50,000 hits) before incrementally raising the target for the entire room based on the energy available in Storage.¹²

Structure	RCL 3 Max Hits	RCL 4 Max Hits	Decay Rate
Wall	300,000,000	300,000,000	None
Rampart	1,000,000	3,000,000	300 / 100 ticks
Road	5,000 (Plains)	5,000 (Plains)	1 per tick (usage)

The Logistics Manager should only authorize "Wall/Rampart Repair" tasks when energy levels in the Storage are healthy (e.g., $> 100,000$), as these are "sinks" that do not provide immediate economic returns.⁵

Imperial Expansion: Global Control and Multi-Room Logic

As the primary room reaches the stability of RCL 4, the system must evaluate its capacity for expansion. Expansion is governed by the Global Control Level (GCL), which is increased by the same upgradeController actions used to level up rooms.³ GCL determines the maximum number of rooms a player can control and also increases the total CPU limit on the official server.³

Expansion Criteria and Site Selection

The "Expansion Planner" or "Strategist" module evaluates potential sites based on a set of criteria designed for long-term sustainability²⁷:

1. **Source Density:** Two-source rooms are significantly more valuable than one-source rooms, as they provide double the energy input (20 e/t vs 10 e/t) for roughly the same defensive and logistic overhead.²⁷
2. **Terrain Accessibility:** Rooms with few swamps are prioritized to reduce movement costs and road maintenance.²³
3. **Remote Mining Potential:** The Strategist checks for nearby neutral rooms that can be

"exploited" through remote mining, which provides an energy-positive return even with sloppy implementations.²⁷

The Colonial Bootstrap Process

When a "Lawyer" (creep with CLAIM parts) successfully claims a new controller, the kernel instantiates a new "Office" or "Colony" object.⁹ The expansion process follows a three-stage bootstrap:

- **Phase 1: Support:** The "Parent Office" sends an "Engineer" to build the first Spawn and a "Paralegal" to maintain the controller's upgrade level.²⁷
- **Phase 2: Transition:** Build requests in the new room are authorized to harvest energy directly from sources to avoid disrupting the parent office's energy pipeline.²⁷
- **Phase 3: Autonomy:** Once the first Spawn is complete, the new room's Logistics Manager takes over, and the room becomes a self-sufficient unit within the hivemind.²⁷

Expansion Minion	Role Name	Body Parts	Objective
Claimer	Lawyer	CLAIM, MOVE	Claim/Reserve Controller
Builder	Engineer	WORK, CARRY, MOVE	Construct first Spawn
Upgrader	Paralegal	WORK, CARRY, MOVE	Prevent room downgrade
Scout	Intern	MOVE	Gain vision and map terrain

The Strategist uses an "Intent" logic to manage rooms: rooms with two sources and good terrain are set to ACQUIRE, while rooms owned by other players are set to AVOID.²⁷

Resource Optimization: CPU and Memory Management

In a kernel-based architecture, CPU is the ultimate constraint. On the official MMO server, players start with a 20 CPU limit, which increases by 10 per GCL level.³ At RCL 3 and 4, the number of creeps and structures grows, necessitating aggressive optimization through the "Analyst" layer.²

Caching Strategies for the Analyst Layer

Analysts are responsible for minimizing redundant API calls by caching data. Caching is divided into categories based on its volatility ²:

- **Static Caching:** Terrain data, source positions, and controller location are cached indefinitely in a room's memory object.
- **Tick Caching:** Creep positions and find results (e.g., FIND_MY_STRUCTURES) are cached for the duration of a single tick to prevent multiple Overlords from making the same expensive calls.²
- **Path Caching:** Pathfinding is the most expensive operation in Screeps.¹³ The AI should store frequently used paths (e.g., Source to Storage) in Memory. Instead of calling creep.moveTo(target), the creep follows the cached path until it reaches the destination or encounter's an unexpected obstacle.³⁰

Intent Economics and Intent Grouping

Every successful action (move, harvest, transfer) has a flat 0.2 CPU cost known as an "Intent".¹³ The Logistics Manager can save CPU by "grouping" actions. For example, instead of a Miner transferring energy to a container every tick, it can wait until its carry capacity is full to perform a single transfer.¹³

Action	CPU Cost (Intent)	Optimization Strategy
creep.move()	0.2	Cache paths, use range: 3 for controllers
creep.harvest()	0.2	Static mining (Miner sits on container)
creep.transfer()	0.2	Batch transfers when full
creep.build()	0.2	Use fewer, larger creeps with more WORK parts

Using a profiler, such as screeps-profiler, is essential for identifying bottlenecks in the kernel. Often, lower-level rooms (RCL 1-3) require more CPU than higher-level rooms because they utilize a larger number of small, manual workers rather than high-capacity specialized units.¹³

Strategic Summary and Operational Outlook

The transition through RCL 3 and RCL 4 is a period of "professionalization" for the AI. The

introduction of the Tower provides the first reliable active defense, but it also creates a new logistics requirement for constant energy replenishment. The attainment of RCL 4 and the subsequent construction of the Storage structure marks the end of the "early game" survival phase, allowing for the accumulation of massive energy buffers that facilitate empire-wide expansion and rapid GCL growth.⁵

For a kernel-based architecture, success at these levels depends on the robust implementation of specialized managers. The Architect must move beyond random placement toward algorithmic efficiency using Distance Transform and Floodfill algorithms.⁸ The Logistics Manager must transition from container-based "energy" properties to storage-based "store" objects, utilizing task-chaining to maximize hauler throughput.⁹ Finally, the Defensive Overlord must prepare for the "Heavy" invader threshold by integrating active counter-squads with tower fire.¹

As the colony progresses toward RCL 5, the kernel should focus on establishing "Remote Mining" operations in adjacent rooms, as this surplus energy is the primary fuel for the next massive jump in RCL requirements. By mastering the logistics and architectural challenges of RCL 4, the AI lays the foundation for advanced technologies such as Links, Terminals, and Lab-based mineral processing that define the late-game experience.

Operational Checklist for RCL 4 Readiness

1. **Architecture:** Ensure the "Architect" has designated a central core for the Storage and a 5x5 stamp for "Fast Fillers" near the spawn.⁸
2. **Logistics:** Update the "Logistics Manager" to prioritize filling the Tower and Extensions from the Storage once it is built.²
3. **Defense:** Deploy a "Guard" directive for the "Defensive Overlord" to spawn combat creeps when "Heavy" invaders (RCL 4+) are detected.⁹
4. **Expansion:** Set the "Strategist" to scout for adjacent two-source rooms with low swamp density for the first expansion once GCL 2 is reached.²⁷
5. **Efficiency:** Enable path caching and intent batching in the "Analyst" layer to keep CPU usage within the 20-30 limit typically found at this stage.¹³

The strategic focus must remain on maximizing energy input—striving for the theoretical maximum of **20 energy/tick**—and leveraging the Storage buffer to smooth out the volatility of the mid-game economy. This approach ensures that the colony remains resilient against NPC threats while maintaining the momentum required to reach the high-utility structures of RCL 5 and beyond.

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